Academic work can & should be embedded in a viable theory of change

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github.com/setgree/presentations

Last year I talked about a meta-analysis

Of interventions intended to reduce consumption of meat and animal products.
Basically nothing works :(



Contents lists available at ScienceDirect

Appetite

journal homepage: www.elsevier.com/locate/appet

Full Length Article

Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis

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ABSTRACT

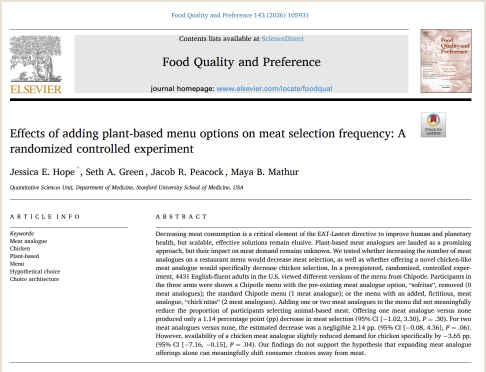
Which interventions produce the largest and most enduring reductions in consumption of meat and animal products (MAP)? We address this question with a theoretical review and meta-analysis of randomized controlled trials that measured MAP consumption at least one day after intervention. We meta-analyze 35 papers comprising 41 studies, 112 interventions, and approximately 87,000 subjects. We find that these papers employ four major strategies to change behavior: choice architecture, persuasion, psychology (manipulating the interpersonal, cognitive, or affective factors associated with eating MAP), and a combination of persuasion and psychology. The pooled effect of all 112 interventions on MAP consumption is quite small (standardized mean difference (SMD) = 0.07 (95 % CI: [0.02, 0.12]), indicating an unsolved problem. Interventions aiming to reduce only consumption of red and processed meat were more effective (SMD = 0.25; 95 % CI: [0.11, 0.38]), but it remains unclear whether such interventions also decrease consumption of other forms of MAP. We conclude that while existing approaches do not provide a proven remedy to MAP consumption, designs and measurement strategies have generally been improving over time, and many promising interventions await rigorous evaluation.

Table 1. Main findings.

Approach	N (Studies)	N (Estimates)	SMD	τ	95% CIs	p val
Overall	41	112	0.07	0.08	[0.02, 0.12]	0.007
Theory						
Choice Architecture	2	3	0.21	0	[-0.99, 1.42]	0.267
Psychology	9	12	0.11	0.03	[-0.03, 0.24]	0.092
Persuasion	25	77	0.07	0.09	[0.01, 0.13]	0.023
Persuasion & Psychology	11	20	0.11	0.19	[-0.06, 0.28]	0.189
Types of Persuasion						
Animal Welfare	16	65	0.03	0.04	[-0.02, 0.07]	0.189
Environment	16	29	0.09	0.12	[-0.02, 0.19]	0.095
Health	18	29	0.08	0.09	[-0.01, 0.17]	0.064

Since then:

TACOS: adding plant-based options to a Chipotle-like menu cut meat by ~1 pp. per option



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Food Quality and Preference

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Effects of adding plant-based menu options on meat selection frequency: A randomized controlled experiment

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Choice architecture

ABSTRACT

Decreasing meat consumption is a critical element of the EAT-Lancet directive to improve human and planetary health, but scalable, effective solutions remain elusive. Plant-based meat analogues are lauded as a promising approach, but their impact on meat demand remains unknown. We tested whether increasing the number of meat analogues on a restaurant menu would decrease meat selection, as well as whether offering a novel chicken-like meat analogue would specifically decrease chicken selection. In a preregistered, randomized, controlled experiment, 4431 English-fluent adults in the U.S. viewed different versions of the menu from Chipotle. Participants in the three arms were shown a Chipotle menu with the pre-existing meat analogue option, "sofritas", removed (0 meat analogues); the standard Chipotle menu (1 meat analogue); or the menu with an added, fictitious, meat analogue, "chick'nitas" (2 meat analogues). Adding one or two meat analogues to the menu did not meaningfully reduce the proportion of participants selecting animal-based meat. Offering one meat analogue versus none produced only a 1.14 percentage point (pp) decrease in meat selection (95% CI [−1.02, 3.30], $P = .30$). For two meat analogues versus none, the estimated decrease was a negligible 2.14 pp. (95% CI [−0.08, 4.36], $P = .06$). However, availability of a chicken meat analogue slightly reduced demand for chicken specifically by −3.65 pp. (95% CI [−7.16, −0.15], $P = .04$). Our findings do not support the hypothesis that expanding meat analogue offerings alone can meaningfully shift consumer choices away from meat.

PALATE: adding alt proteins to seven real restaurants did not reduce meat purchases



New alternatives, same orders: modern alternative proteins did not replace meat purchases in restaurants

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Abstract

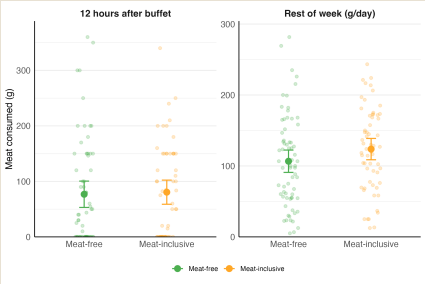
Plant-based alternative proteins are widely expected to displace meat consumption and, in turn, reorient the global food system away from the externalities of animal agriculture, but little empirical work has assessed these presumed substitution effects in naturalistic settings. Here, we use a dataset of 1.2 million transactions across seven restaurants and eight years to investigate whether introducing modern alternative proteins to restaurant menus reduces purchases of meat-containing dishes. We analyzed effects on total meat purchases and on category-matched purchases, pairing each alternative protein with the meat it emulates; we also estimate individual (within-customer) purchasing changes. Our results from 40 preregistered outcome-exposure pairings are overwhelmingly null, whether pooled across restaurants or stratified by restaurant. These findings challenge the widespread belief that alternative proteins displace meat purchases.

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Studies I'm working on now

- Does a plant-based lunch cause rebound/reinforcement effects?
  - No
- Does telling people about the small animal replacement problem make them endorse paying more for higher welfare meat?
  - IDK
  - BTW Brian this is definitely reducetarian in spirit because higher welfare -> higher prices -> lower welfare
- W/ Greener by Default: which plant-based default works best?
  - IDK

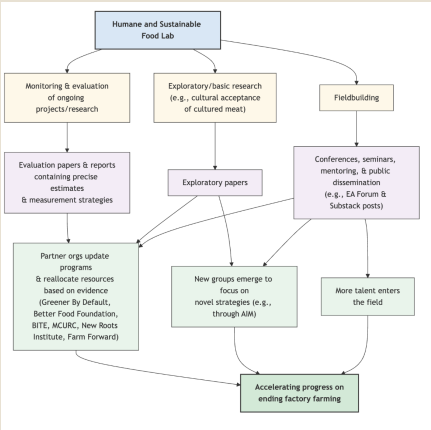


My theory of change (why I do this work)

1. I try to discover true facts that will be useful and interesting
   - Animal advocacy has an evidence problem across many domains
   - Donors build these big complicated CEA models on weak foundations; I want to improve those foundations
2. A lot of papers are like 'this one weird trick will solve [pressing social problem]' and I live to fight them
   - The point of https://regressiontothemeat.substack.com/ is basically "you should not take (some piece of) research at face value because [publication bias/misleading presentation/mismatch between strength of evidence & strength of claim/etc.]"
3. If the advocates do their own evaluations they'll all say how great they are

My *lab's* theory of change

I actually had to think about this for a fundraising thing (inputs, outputs, outcomes, and impact):



My take: if you don't articulate your TOC and design your research around it, you might default to a TOC that makes no sense

1. 'Effect change within my field'
   - This is fine if you can pull it off but it's A LOT OF WORK. Talk to Kristof/Kevin Kuruc/Bob Fischer about this
2. Join the vegan echo chamber
   - Write papers like 'carnists score higher on sociopathy/RWA measures' or 'it is ethically justified to serve your colleagues a vegan meal'
3. 'My thing definitely works and everyone should invest in it'
   - Once you *know* the answer to the big questions, it's really hard to do research because you're no longer searching
   - Aesthetic objection: this profanes our sacred duty to seek truth
   - Just go get an advocacy job instead

So what's your TOC?

- The fact that I don't understand what you do is neither here nor there (who cares what I think)
- If you want someone to take your work and run with it, what does that look like?
  - Who would it be?
- If you think your question is interesting, what is it a case of?
  - What broader phenomenon are you hoping to illuminate?
- Thanks!

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